Carboxylic Acids

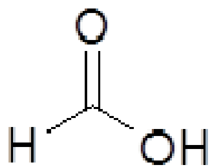
CHE 1203 – Organic Chemistry 1

The Importance of Carboxylic Acids (RCO_2H)

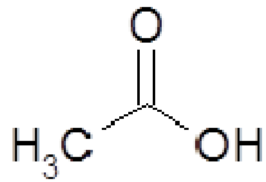
- Abundant in nature from oxidation of aldehydes and alcohols in metabolism
  - Acetic acid (CH_3CO_2H) - vinegar
  - Butanoic acid ($CH_3CH_2CH_2CO_2H$) – smell of rancid butter
  - Caproic acid ($CH_3CH_2CH_2CH_2CH_2CO_2H$) - smell of sweaty gym socks
  - Long-chain aliphatic acids from the breakdown of fats

- Starting materials for *acyl derivatives* (esters, amides, and acid chlorides)

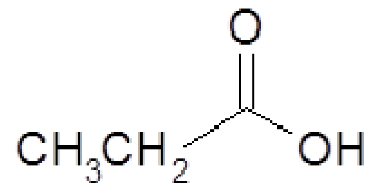
Some Common Names



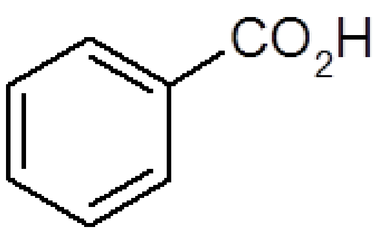
Formic



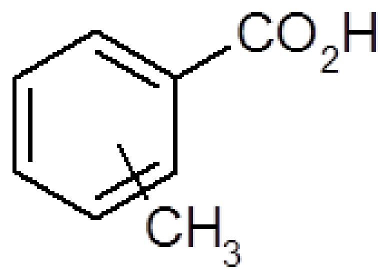
Acetic



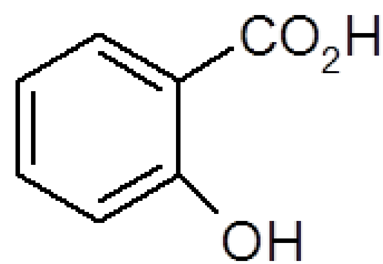
Propionic



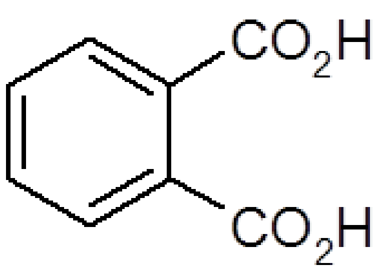
Benzoic



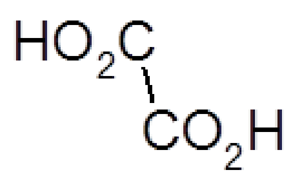
Toluic



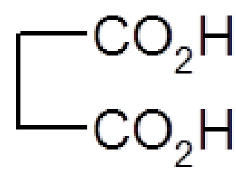
Salicylic



Phthalic



Oxalic

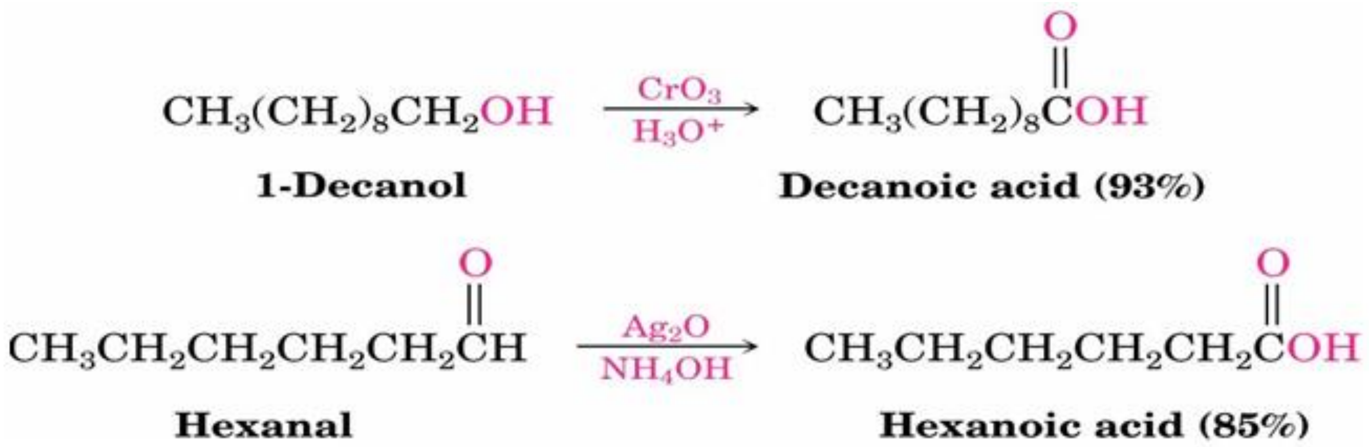


Succinic

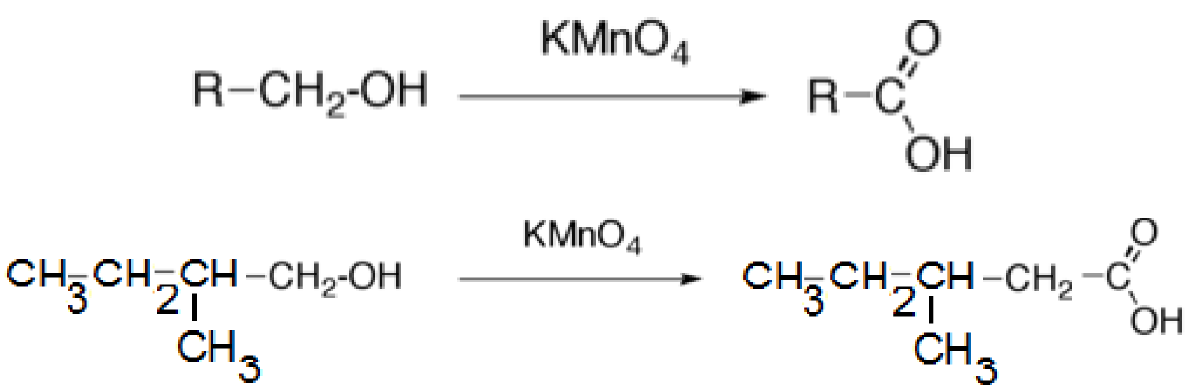
Preparation of Carboxylic Acids

From Alcohols

- Oxidation of a primary alcohol or an aldehyde with CrO_3 in aqueous acid



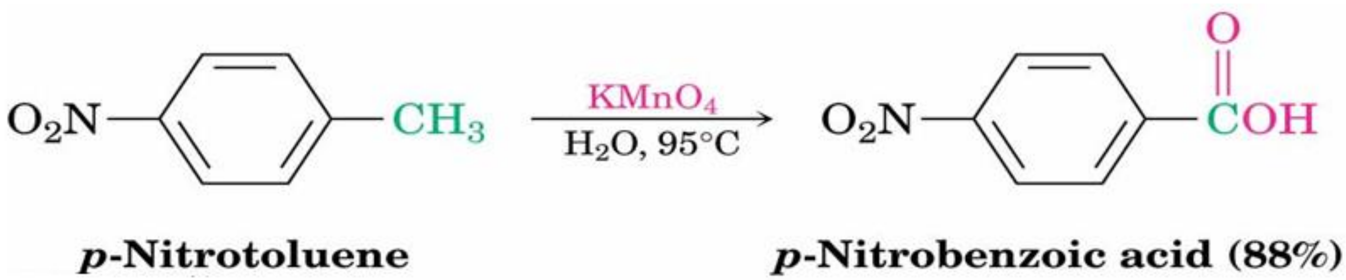
- Oxidation of a primary alcohol with $KMnO_4$



1. Oxidation of alkylbenzene

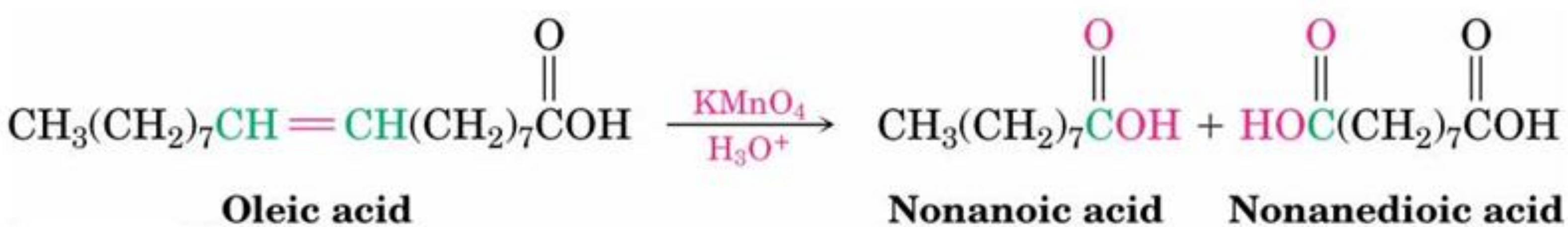
Oxidation of a substituted alkylbenzene with $KMnO_4$ or $Na_2Cr_2O_7$ gives a substituted benzoic acid

1° and 2° alkyl groups can be oxidized, but tertiary groups do not react



2. From Alkenes

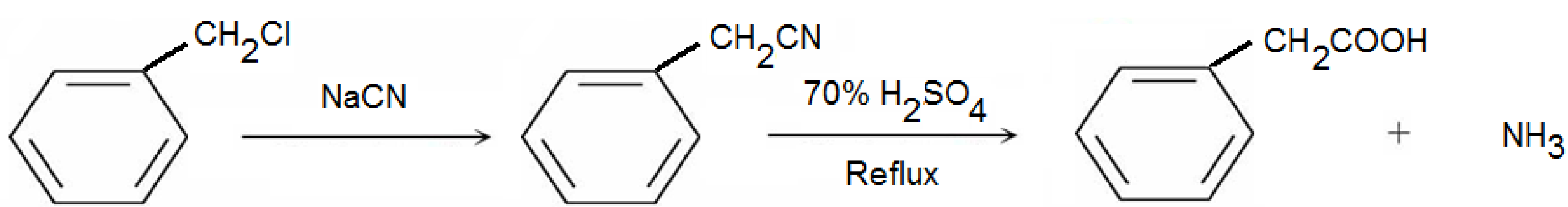
- Oxidative cleavage of an alkene with $KMnO_4$ gives a carboxylic acid if the alkene has at least one vinylic hydrogen



Hydrolysis of Nitriles

- Hot acid or base yields carboxylic acids
- Conversion of an alkyl halide to a nitrile (with cyanide ion) followed by hydrolysis produces a carboxylic acid with one more carbon
- $RBr \rightarrow RC{\equiv}N \rightarrow RCO_2H$
- Best with primary halides because elimination reactions occur with secondary or tertiary alkyl halides

$$RCH_2Br \xrightarrow[(S_N2)]{Na^+ \ ^-CN} RCH_2C{\equiv}N \xrightarrow{H_3O^+} RCH_2\overset{\displaystyle O}{\overset{\|}{C}}OH + NH_3$$



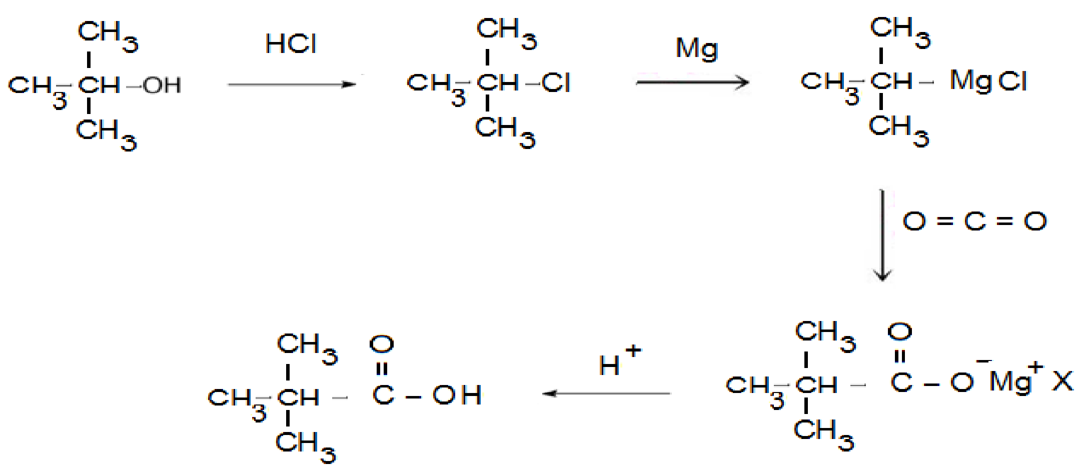
Carboxylation of Grignard Reagents

- Grignard reagents react with dry CO_2 to yield a carboxylic acid after acid hydrolysis
- Limited to alkyl halides that can form Grignard reagents

$$R\text{-}X \xrightarrow{Mg} R\text{-}MgX \xrightarrow{CO_2} R\text{--}COO^{-}Mg^{+}X \xrightarrow{H+} R\text{-}COOH$$

1-Bromo-2,4,6-trimethylbenzene

2,4,6-Trimethylbenzoic acid (87%)



Reactions of Carboxylic Acids

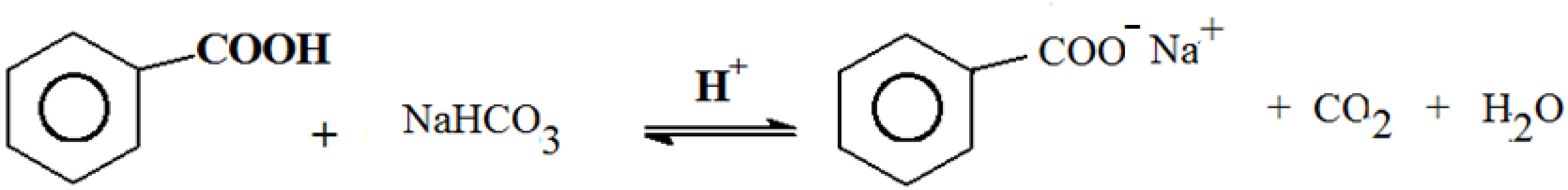
- Reactions of acids are consist of those due to a). Hydroxyl group b). R group c). Carbonyl d) Carbonyl group as a whole
- Carboxylic acids transfer a proton to a base to give carboxylic anions, which are good nucleophiles in S_N2 reactions
- Like ketones, carboxylic acids undergo addition of nucleophiles to the carbonyl group
- In addition, carboxylic acids undergo other reactions characteristic of neither alcohols nor ketones

Reactions due to Hydroxyl group

1. Salt formation

$$R-COOH \rightleftharpoons RCOO^- + H^+$$

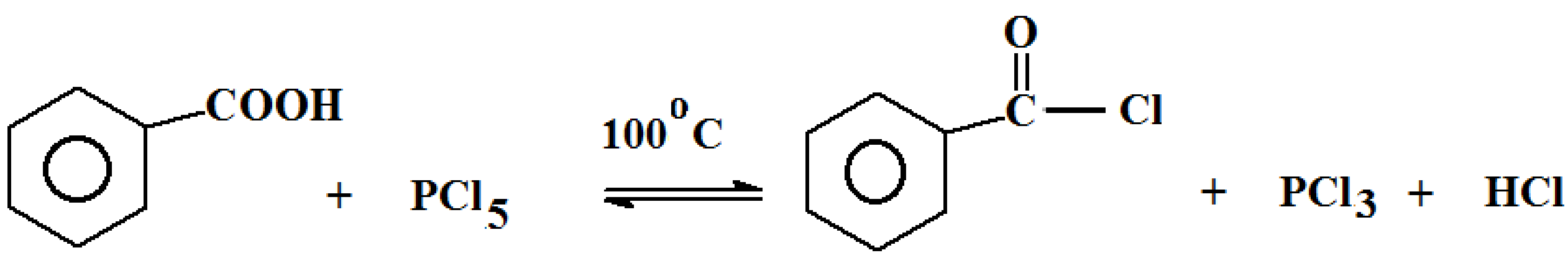
$$2\,CH_3COOH + Zn \qquad (CH_3CO\bar{O})_2\,Zn^{2+} + H_2$$



Release of CO_2 is visible, used as a test for presence of -COOH group

Conversion into acid chlorides

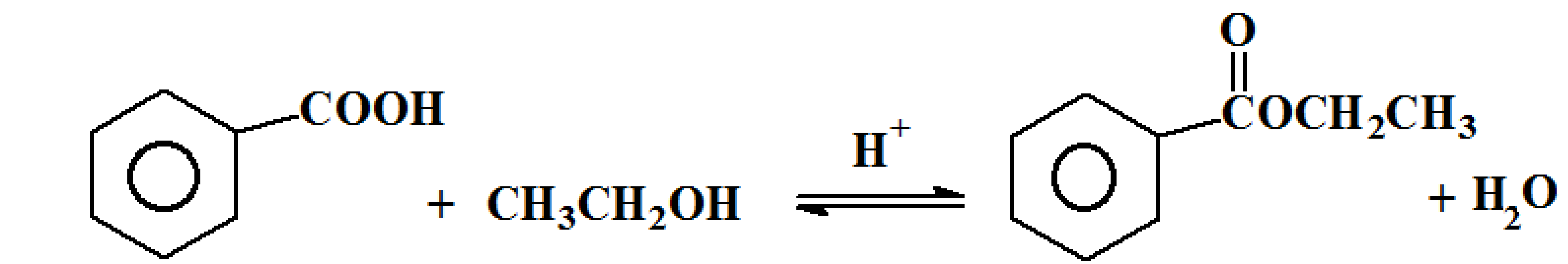
$$\text{R}-\text{COOH} + \begin{matrix}\text{SOCl}_2 \\ \text{PCl}_3 \\ \text{PCl}_5\end{matrix} \longrightarrow \text{RCOCl}$$

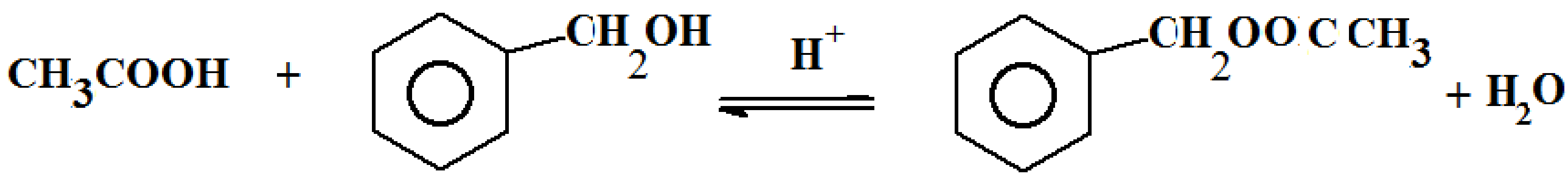


$$n\text{-}C_{17}H_{35}COOH + SOCl_2 \xrightarrow{\text{reflux}} n\text{-}C_{17}H_{35}COCl + SO_2 + HCl$$

Esterification

- Acid + alcohol yields ester + water.
- Acid catalyzed for weak nucleophile.
- All steps are reversible.
- Reaction reaches equilibrium.   Rate $1^{ry} > 2^{ry} > 3^{ry}$





$$R-\overset{\overset{\large O}{\|}}{C}-OH \xrightarrow{SOCl_2} R-\overset{\overset{\large O}{\|}}{C}-Cl \xrightarrow{ROH} R-\overset{\overset{\large O}{\|}}{C}-OR$$

Formation of Anhydrides

- Acids when treated with a strong dehydrating agent, eliminate a molecule of water giving anhydrides of acids

$$R-\overset{\overset{O}{\|}}{C}-OH \quad + \quad R-\overset{\overset{O}{\|}}{C}-OH \xrightarrow[-\,H_2O]{P_2O_5} R-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-R$$

Formation of Amides

- Acids when treated with ammonia first and then heating result amide

$$R-\overset{\overset{O}{\|}}{C}-OH \quad + \quad NH_2OH \longrightarrow \underset{\text{Ammonium salt}}{RCO_2\,NH_4} \xrightarrow[\blacktriangle]{} \underset{\text{amide}}{RCONH_2} \quad + \quad H_2O$$

$$\underset{\text{Ammonium salt}}{2\ RCO_2\,NH_4} \xrightarrow[\text{heat}]{P_2O_5} \underset{\text{Alkyl nitrile or alkyl cyanide}}{RCN} \quad + \quad 2\ H_2O$$

$$R-\overset{\overset{O}{\|}}{C}-OH \xrightarrow{SOCl_2} R-\overset{\overset{O}{\|}}{C}-Cl \xrightarrow{NH_3} R-\overset{\overset{O}{\|}}{C}-NH_2$$

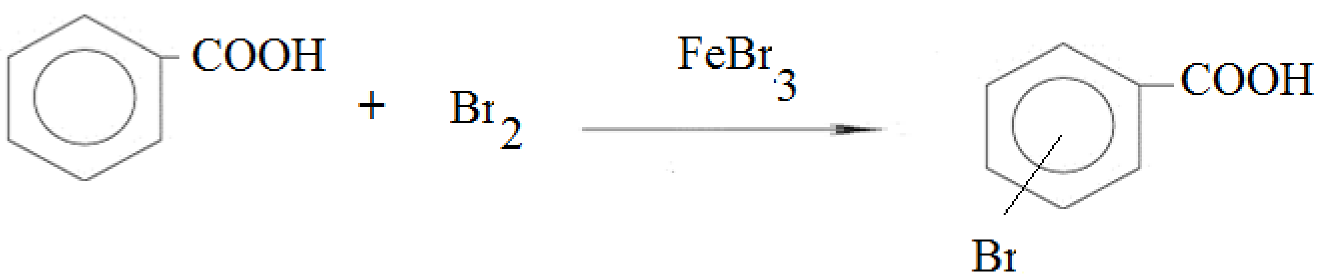
b). Reactions due to R group

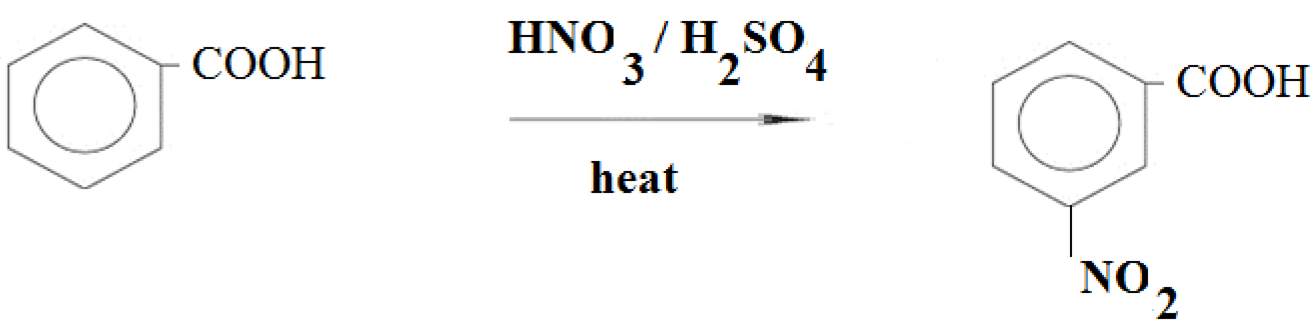
- In the presence of catalyst such as Iodine or Phosphorus, Chlorine and Bromine attack the aliphatic hydrocarbon group, R, forming substituted acids.

$$CH_3COOH + Cl_2 \xrightarrow{I_2 \text{ or } P} Cl{-}CH_2COOH + HCl$$

Acetic acid monochloro acetic acid

- Aromatic hydrocarbon group react with Chlorine or Bromine in the presence of Ferric chloride or Ferric bromide as a catalyst.

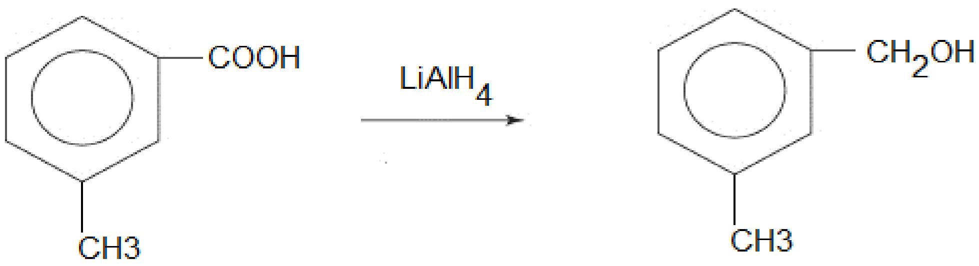




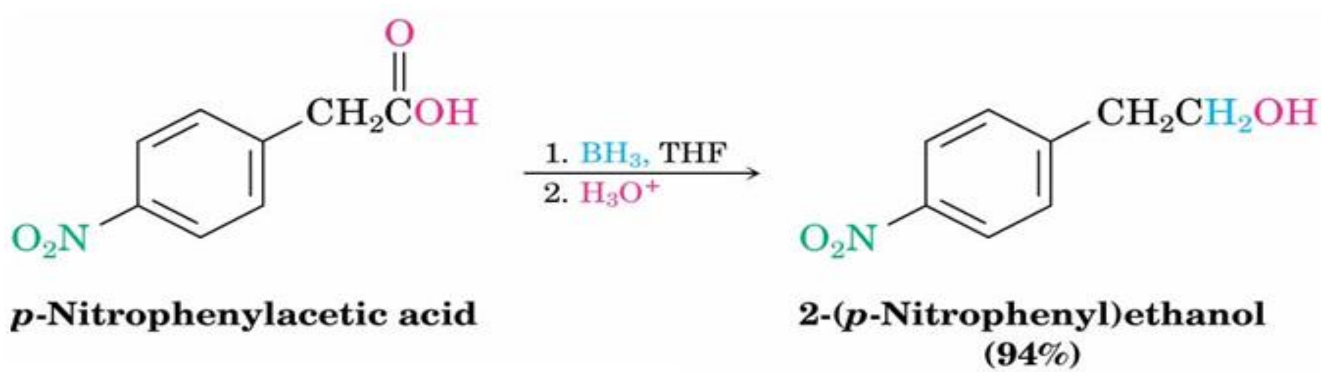
c). Reactions due to carbonyl group :

Reduction of Carboxylic Acids

- Reduced by $LiAlH_4$ to yield primary alcohols
- The reaction is difficult in lower fatty acids
- Generally requires heating in THF



- Borane in tetrahydrofuran (BH_3/THF) converts carboxylic acids to primary alcohols selectively
- Preferable to $LiAlH_4$ because of its relative ease, safety, and specificity



d). Reactions of acids due to Carbonyl group as a whole

Decarboxylation

$$R-\overset{\overset{\large O}{\|}}{C}-OH \xrightarrow[\text{heat}]{\text{Soda- lime}} RH + CO_2$$

Electrolytic Decarboxylation

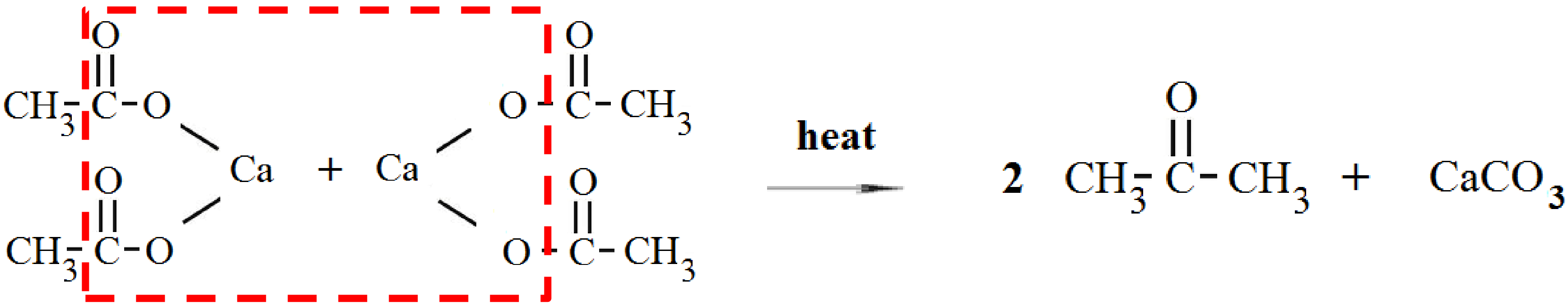
$$2\ RCOONa \xrightarrow[2\ H_2O]{\text{electrolysis}} R-R + 2\ CO_2 + 2\ NAOH + H_2$$

Formation of carbonyl compounds

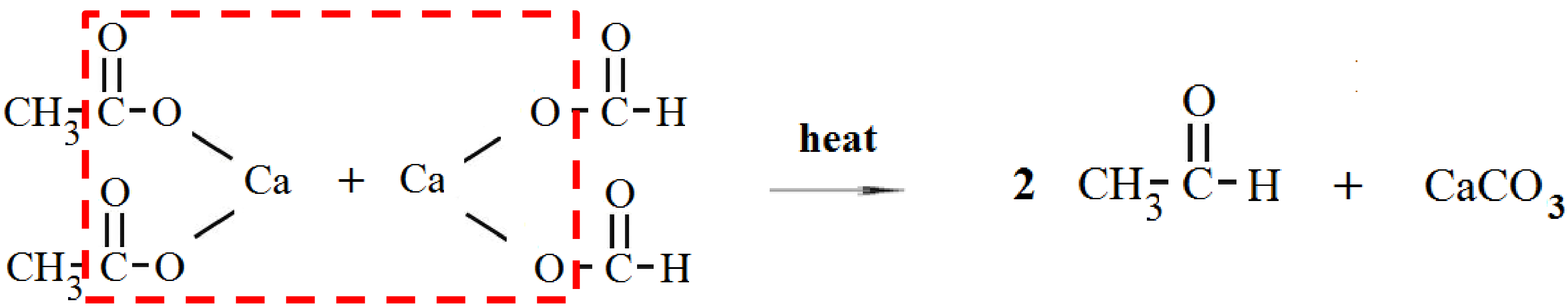
$$2\ (CH_3COO)_2Ca \xrightarrow{\text{heat}} (CH_3)_2C=O + CaCO_3$$

Calcium acetate — Acetone

Formation of carbonyl compounds……….



2 Calcium acetate Acetone



Calcium acetate Calcium formate Acetaldehyde

Nitriles

- Nitriles and carboxylic acids both have a carbon atom with three bonds to an electronegative atom, and both contain a π bond
- Both are electrophilic centers

A nitrile—three bonds to nitrogen

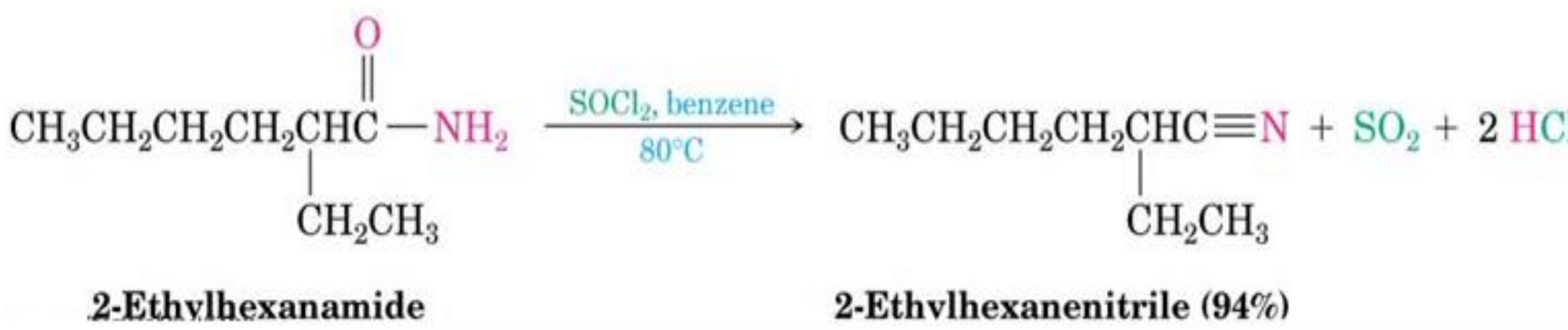
An acid—three bonds to two oxygens

Preparation of Nitriles

- Primary alkyl halides + cyanide
- S_N2 reaction – same constraints

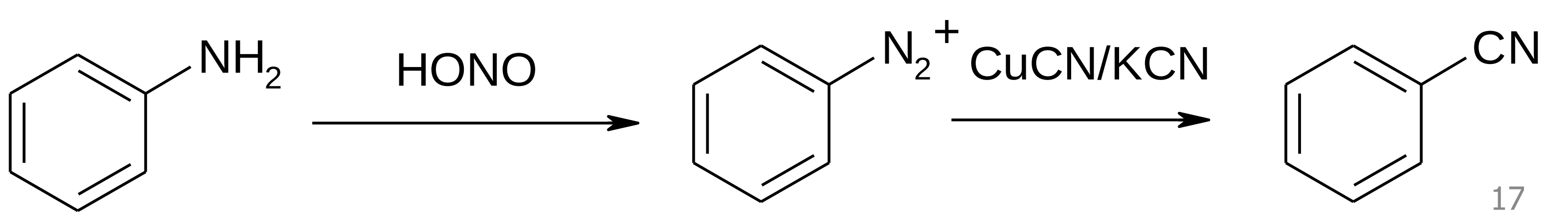
$$R{-}X \xrightarrow{KCN} RCN$$

- Primary amides $RCONH_2$ react with $SOCl_2$ or $POCl_3$ (or other dehydrating agents)
- Not limited by steric hindrance or side reactions (as is the reaction of alkyl halides with NaCN)



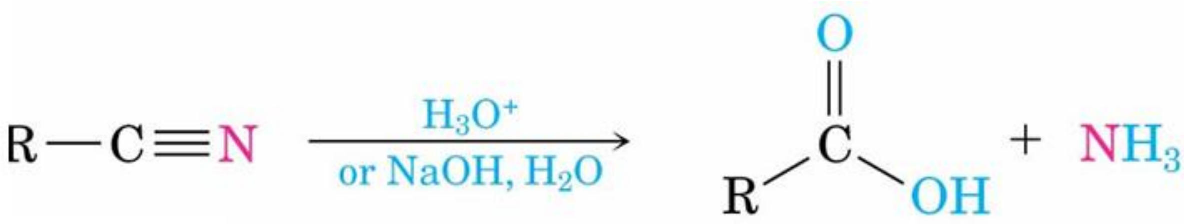
Aromatic Nitriles

- Conversion of anilines into diazonium salt
- Followed by CuCN/KCN



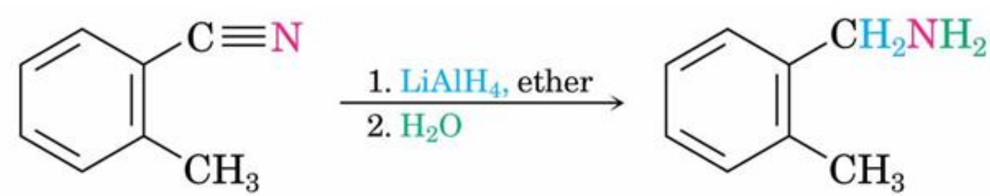
Hydrolysis of Nitriles

- Hydrolyzed by acid or base to the carboxylic acid
- Ammonia or an amine is other product



Reduction of Nitriles

- Reduction of a nitrile with $LiAlH_4$ gives a primary amine



***o*-Methylbenzonitrile** **_o_-Methylbenzylamine (88%)**

Nitriles with Organometallic Reagents

- Grignard reagents add to give an intermediate imine anion that is hydrolyzed by addition of water to yield a ketone